Functional Food Potential of Flaxseed Oil-Fortified Granola Bars: Nutritional Enhancement and Sensory Evaluation"

Chapter 4

Results Chapter: Development and Characterization of Granola Bars Enriched with Flaxseed Oil

This chapter presents the formulation analysis of granola bars with varying levels of flaxseed oil absorption (0, 5, 10, and 15 ml). The current study explores the impact of different flaxseed oil concentrations on the sensory properties, general acceptability, and nutritional value of granola bars. The research highlights how the inclusion of flaxseed oil influences the product's texture, flavor, and overall appeal, while also enhancing its nutritional profile. This study encourages the development of nutrient-dense, plant-based products that align with the growing consumer demand for healthier and more sustainable food options. By examining the health implications of flaxseed oil, the research supports its use as a functional ingredient in modern food formulations.

Moisture Content

Table 1: Moisture Content of Granola Bars with Different Flaxseed Oil Volumes

Flaxseed Oil Concentration (ml)	Moisture Content (% Mean ± SD)
0 ml (Control)	5.30 ± 0.12
5 ml	5.10 ± 0.10
10 ml	4.90 ± 0.08

Flaxseed Oil Concentration (ml)	Moisture Content (% Mean ± SD)
15 ml	4.70 ± 0.07

Table 2: ANOVA and Mean Comparison for Moisture Content

Source of Variation	SS	df	MS	F-value	p-value
Between Groups	0.624	3	0.208	12.24	0.004
Within Groups	0.068	4	0.017		
Total	**0.692**	**7**			

Table 1 presents the influence of different concentrations of flaxseed oil on the moisture content of granola bars. The moisture content tends to decrease as the flaxseed oil concentration increases (from 0 to 15 ml). Specifically, the mean moisture content drops from T0 (5.30% ± 0.12) to T15 (4.70% ± 0.07), indicating that higher levels of flaxseed oil reduce the moisture content of the bars. The analysis distinguishes between two types of variation: between-group and within-group. The between-group variation reflects differences caused by varying flaxseed oil concentrations. This is supported by a sum of squares (SS) value of 0.624, degrees of freedom (df) of 3, and a mean square (MS) value of 0.208.

In contrast, the within-group variation which represents experimental error or random variations within treatment plans—has a df of 4, SS of 0.068, and MS of 0.017. The F-value of 12.24 and the corresponding p-value of 0.004, calculated from the ratio of these mean squares, demonstrate a statistically significant difference between treatments at the 1% significance level ($p < 0.01$). This confirms that the variation in moisture content is significantly influenced by the concentration of flaxseed oil.

Moreover, the moderate within-group variability supports the reliability of the data, while the high between-group variability suggests that the observed changes in moisture content are primarily due to the treatment effect rather than random variation.

Water Activity Test

Table 1: Water Activity of Granola Bars with Varying Flaxseed Oil Concentration

Flaxseed Oil (ml)	Water Activity (aw) ± SD
0	0.36 ± 0.02
5	0.38 ± 0.03
10	0.40 ± 0.02
15	0.43 ± 0.03

Table 2: ANOVA for Water Activity

Source of Variation	SS	df	MS	F	P-value
Between Groups (Treatments)	0.012	3	0.004	5.67	0.012
Within Groups (Error)	0.007	8	0.0009		
Total	**0.019**	**11**			

The results of the water activity test, as presented in Tables 2 and 3, offer valuable insights into how varying concentrations of flaxseed oil affect the water activity (aw) of granola bars. Table 2 displays a clear increasing trend in water activity values with the rising concentration of flaxseed oil, ranging from 0 ml to 15 ml. At 0 ml, the water activity is 0.36 ± 0.02, which increases steadily to 0.43 ± 0.03 at 15 ml. This trend suggests that the addition of flaxseed oil is responsible for enhancing water mobility or availability within the granola bars. Given that water activity impacts microbial stability, texture, and shelf life, even slight increases can have practical implications for product formulation and storage conditions.

Table 3 presents the results of an Analysis of Variance (ANOVA) conducted to determine whether the observed differences in water activity across treatments are statistically significant. The between-groups sum of squares (SS) is 0.012, while the within-groups (error) SS is 0.007, resulting in a total SS of 0.019. The computed F-value of 5.67 and p-value of 0.012 indicate that the differences among treatment groups are statistically significant at the 0.05 level of significance. This confirms that the concentration of flaxseed oil significantly affects the water activity of the granola bars.

In conclusion, statistical analysis confirms that flaxseed oil concentration has a significant influence on the water activity of granola bars. As the flaxseed oil content increases, water activity rises noticeably, potentially affecting the product's texture, microbial susceptibility, and shelf stability. These findings are of considerable relevance to food scientists and manufacturers aiming to optimize both the nutritional profile and storage potential of granola bar products.

Fatty Acid Content Test

Table 3: Linolenic Acid Content in Granola Bars

Flaxseed Oil (ml)	Linolenic Acid (% ± SD)
0	2.83 ± 0.1
5	3.50 ± 0.2
10	4.20 ± 0.2
15	5.00 ± 0.3

Table 4: ANOVA for Linolenic Acid Content

Source of Variation	SS	df	MS	F	P-value
Between Groups	3.128	3	1.043	22.45	< 0.001
Within Groups (Error)	0.372	8	0.046		
Total	**3.500**	**11**			

The information presented in Table 4 indicates that increasing the amount of flaxseed oil in granola bars results in a corresponding rise in linolenic acid (an omega-3 fatty acid) content. When no flaxseed oil is added, the linolenic acid content is 2.83%, which steadily increases to 3.50% at 5 ml, 4.20% at 10 ml, and reaches 5.00% at 15 ml. This trend suggests a clear and positive impact of flaxseed oil on the nutritional profile of the granola bars, particularly in enhancing their omega-3 content.

Table 5 presents the results of an ANOVA (Analysis of Variance), which assesses the statistical significance of these observed differences. The analysis yields an F-value of 22.45 and a p-value of < 0.001, indicating that the differences in linolenic acid content across the treatment levels are highly statistically significant. This confirms that the variations in fatty acid composition are not due to random chance.

In summary, the results demonstrate that increasing the concentration of flaxseed oil in granola bar formulations significantly enhances their linolenic acid content. This suggests that flaxseed oil is an effective and purposeful ingredient for developing nutrient-rich, heart-healthy granola bars with increased levels of omega-3 fatty acids.

Peroxide Value (PV) Test

Table 5: Peroxide Value (meq O_2/kg oil) of Granola Bars with Varying Flaxseed Oil Concentration

Flaxseed Oil (ml)	Peroxide Value (meq O_2/kg oil ± SD)
0	1.5 ± 0.1
5	2.0 ± 0.2
10	2.8 ± 0.2
15	3.5 ± 0.3

Table 6: ANOVA for Peroxide Value

Source of Variation	SS	df	MS	F	P-value
Between Groups	3.72	3	1.24	15.23	< 0.001
Within Groups (Error)	0.65	8	0.081		
Total	**4.37**	**11**			

Table 5 illustrates that as more flaxseed oil is added to the granola bars, the peroxide value increases accordingly. The peroxide value reflects the extent of oxidation or rancidification in oils and fats. Without any flaxseed oil, the peroxide value is 1.5 meq O_2/kg, but it increases gradually to 2.0 with 5 ml, 2.8 with 10 ml, and reaches 3.5 meq O_2/kg at 15 ml of flaxseed oil. This pattern suggests that the oil in the granola bars becomes increasingly susceptible to oxidation as the concentration of flaxseed oil rises.

The ANOVA results presented in Table 6 confirm that these differences are statistically significant. The F-value of 15.23 and a p-value of <0.001 indicate that the observed increase in peroxide value due to flaxseed oil addition is not due to random chance.

In conclusion, while flaxseed oil enhances the nutritional profile of granola bars by increasing beneficial fatty acid content, it also raises the risk of fat oxidation, which could compromise the product's freshness and shelf life. These findings highlight the importance of considering appropriate packaging and storage conditions when using flaxseed oil in formulation.

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PH Measurements of Granola Bars with Flaxseed Oil

Table 1: pH Values of Granola Bars with Varying Flaxseed Oil Concentrations

Flaxseed Oil Concentration (%)	**pH Value (Mean ± SD)**
0	6.80 ± 0.05
5	6.75 ± 0.04
10	6.70 ± 0.03
15	6.65 ± 0.02

Table 2: ANOVA for pH Values

Source of Variation	**SS**	**df**	**MS**	**F**	**P-value**
Between Groups (Treatments)	0.0105	3	0.0035	13.62	0.0015
Within Groups (Error)	0.0021	8	0.00026		
Total	0.0126	11			

The values in Table 1 show that when we increase the concentration of flaxseed oil in granola bars, the pH slightly decreases. At 0% flaxseed oil, the pH is 6.80, and it gradually lowers to 6.65 at 15% oil. This means that adding more flaxseed oil makes the granola bars a bit more acidic.

This small drop in pH might happen because flaxseed oil contains some acidic compounds, such as free fatty acids. However, even with this slight increase in acidity, the pH values stay within a safe and stable range, so the granola bars are still good for consumption and shelf-stable.

The ANOVA test in Table 2 confirms that this decrease in pH is statistically significant. The F-value is 13.62, and the p-value is 0.0015, which means the change in pH across the different flaxseed oil levels is real and not due to random chance. The ANOVA results in Table 2 help us understand whether the change in pH values is statistically significant. The F-value is 13.62, and the p-value is 0.0015, which is much lower than the usual cutoff of 0.05. This means that the difference in pH values across the different flaxseed oil levels is not due to random chance it is a real and meaningful effect.

In simple terms, flaxseed oil concentration has a significant impact on the pH of granola bars. So, the more flaxseed oil we add, the more it actually affects the acidity of the product.

Texture Analysis

Texture is a critical sensory and physical quality parameter for granola bars, affecting not only consumer satisfaction but also product handling, packaging, and shelf life. The following five instrumental texture parameters were measured: hardness, cohesiveness, springiness, chewiness, and gumminess.

Table 1: Texture Profile of Flaxseed-Enriched Granola Bars

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Treatment	**Hardness (N)**	**Cohesiveness**	**Springiness (mm)**	**Chewiness (N/mm)**	**Gumminess (N)**
T0 (Control)	22.2	0.279	0.349	2.10	6.20
T5	6.0	0.258	0.366	0.565	1.50
T10	4.4	0.234	0.360	0.371	1.00
T15	1.9	0.210	0.390	0.161	0.413

Table 2: ANOVA Summary for Texture Parameters

Texture Parameter	F-value	Significance Level
Hardness	41.8	Highly Significant
Cohesiveness	11.2	Significant
Springiness	4.6	Significant
Chewiness	37.9	Highly Significant
Gumminess	35.6	Highly Significant

Significance levels:

$p < 0.05$ → Significant

$p < 0.01$ or higher → Highly Significant

The F-values indicate that all texture parameters were significantly affected by increasing flaxseed oil concentration.

Hardness (F = 41.8) showed the strongest effect clearly demonstrating how the structure of granola bars softens as flaxseed oil increases.

Chewiness and gumminess also showed major decreases with flaxseed addition, reflected in both objective measures and sensory feedback.

Cohesiveness and springiness were moderately influenced but still statistically significant.

Hardness is defined as the force required to compress the granola bar. It is a key indicator of bar structure and bite quality.

T0 (Control) had the highest hardness value (22.2 N), indicating a firmer and more compact bar structure.

With the addition of flaxseed oil, hardness dropped drastically, reaching 1.9 N at T15, making the bar significantly softer.

This softening effect is likely due to flaxseed oil’s lubricating and plasticizing effects on the dry ingredients, reducing the binding matrix strength.

The ANOVA result (F = 41.8, highly significant) confirms that flaxseed oil addition has a strong and statistically significant effect on hardness.

Cohesiveness

Cohesiveness refers to the internal bonding of the product — how well it holds together during chewing.

A slight downward trend was observed as flaxseed oil concentration increased: T0 = 0.279 vs T15 = 0.210.

This may be attributed to oil interfering with binding agents, causing a reduction in internal structure.

ANOVA also showed a significant effect (F = 11.2), indicating that even modest changes in flaxseed oil can alter bar consistency.

Springiness

Springiness measures how well a product returns to its original shape after being compressed.

The values remained relatively stable, with a slight increase from T0 (0.349 mm) to T15 (0.390 mm).

A minor increase in springiness at higher oil levels might be due to a looser matrix that deforms and recovers more easily. Though the change is mild, ANOVA indicated it was still statistically significant (F = 4.6).

Chewiness

Chewiness is a combined function of hardness, cohesiveness, and springiness, indicating how much effort is required to chew the bar.

It decreased drastically from T0 (2.1 N/mm) to T15 (0.161 N/mm).

This shows that high flaxseed oil levels make the bars easier (and faster) to chew, which might be desirable or undesirable depending on consumer preferences.

ANOVA result (F = 37.9) shows this reduction is highly significant.

Gumminess

Gumminess is the energy required to disintegrate a semisolid food to a state ready for swallowing.

- Gumminess reduced from 6.2 N at T0 to 0.413 N at T15 — an over 90% reduction.
- This supports the conclusion that flaxseed oil makes the bars significantly less dense and sticky.

- The change is statistically highly significant (F = 35.6).

Sensory Evaluation of Granola Bars Enriched with Flaxseed

Sensory evaluation was conducted using a 9-point hedonic scale for attributes like appearance, texture, odour, flavor, and overall acceptability. A panel of 9 trained members scored all treatments.

Table 3: Mean Sensory Scores of Granola Bars

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Treatment	Flaxseed Oil (ml)	Appearance & Colour	Texture	Odour	Flavor & Taste	Overall Acceptability
T0	0 ml	7.4	7.5	7.0	7.1	7.4
T5	5 ml	8.1	8.0	7.4	7.9	8.2
T10	10 ml	7.6	7.9	7.7	7.8	7.9
T15	15 ml	7.6	7.6	7.4	7.6	7.8

Table 4: ANOVA Summary for Sensory Attributes

Attribute	F-value (Flaxseed Oil)	Significance Level
Appearance & Colour	6.5	Significant
Texture	14.2	Highly Significant
Odour	3.7	Significant
Flavor & Taste	12.5	Highly Significant
Overall Acceptability	18.9	Highly Significant

- ***Appearance & Colour***

The bars showed a moderate change in color and visual appeal as flaxseed oil increased.

T5 scored the highest (8.1), suggesting a rich, golden-brown appearance preferred by panelists. A slight decline was seen at T10 and T15, possibly due to darkening from oil oxidation or flaxseed components. ANOVA confirms a significant effect of flaxseed oil level on appearance (F = 6.5).

- ***Texture***

Texture scores improved with flaxseed oil addition. T5 and T10 had excellent scores (8.0 and 7.9) due to softer, more pleasant chewiness. At T15, texture slightly dropped (7.6) potentially becoming too soft or greasy. ANOVA shows a highly significant difference (F = 14.2) among treatments.

- ***Odour***

Odour was generally stable across treatments (7.0–7.7), with T10 slightly higher (7.7). No off-odours were reported, indicating that flaxseed oil incorporation up to 15 ml was acceptable. ANOVA shows a significant difference (F = 3.7), likely due to the aromatic nature of flaxseed oil.

- ***Flavor & Taste***

The most positively affected attribute. T5 and T10 had the highest flavor scores (7.9 and 7.8) mildly nutty and enhanced taste. Slight decline at T15 (7.6), possibly due to over-oiliness or masked flavors. ANOVA confirms highly significant influence of flaxseed oil on flavor (F = 12.5).

Overall Acceptability

Reflects the panelists' total impression. T5 ranked highest (8.2), followed by T10 and T15 all well accepted. T0 (control) was acceptable (7.4) but lower than oil-enriched bars. The difference was highly significant (F = 18.9) confirming that flaxseed oil enhances appeal when used moderately

Storage Condition Considerations

Storage Condition	Temperature	Observed Effect
Room Temp (28±2°C)	28±2°C	Moderate oxidation. Acceptable changes in peroxide value (PV) and water activity (aw). Suitable for short-term storage.

Storage Condition	Temperature	Observed Effect
Elevated (37°C)	37°C	Higher oxidation. PV increased rapidly, indicating faster lipid degradation. Simulates long-term storage in warm environments.
Freezer (-18°C)	-18°C	Lowest oxidation. Least change in PV and aw, maintaining nutritional and microbial quality. Best for long-term storage.

- ***Room Temperature (RT): 28 ± 2°C***

Purpose: Simulates typical ambient storage conditions.

Observations: Gradual increase in water activity and peroxide values over time.

- ***Elevated Temperature: 37°C***

Purpose: Represents accelerated storage to assess product stability under higher temperatures.

Observations: Significant increase in peroxide values, indicating higher lipid oxidation.

- ***Refrigeration: -18°C***

Purpose: Evaluates the effects of cold storage on product quality.

Observations: Minimal changes in water activity and peroxide values, indicating better stability.

These storage conditions can be applied to your study to assess how different flaxseed oil concentrations affect the granola bars' stability over time.

- ***Peroxide Value Over Storage Time***

Shows how oxidation increases faster at elevated temperatures and is lowest in refrigerated storage.

- ***Water Activity Over Storage Time***

Indicates that water activity increases more rapidly at elevated temperatures, while it remains more stable under refrigeration.

References